

Data Management Plans (DMPs)

Introduction to Data Management Practices course

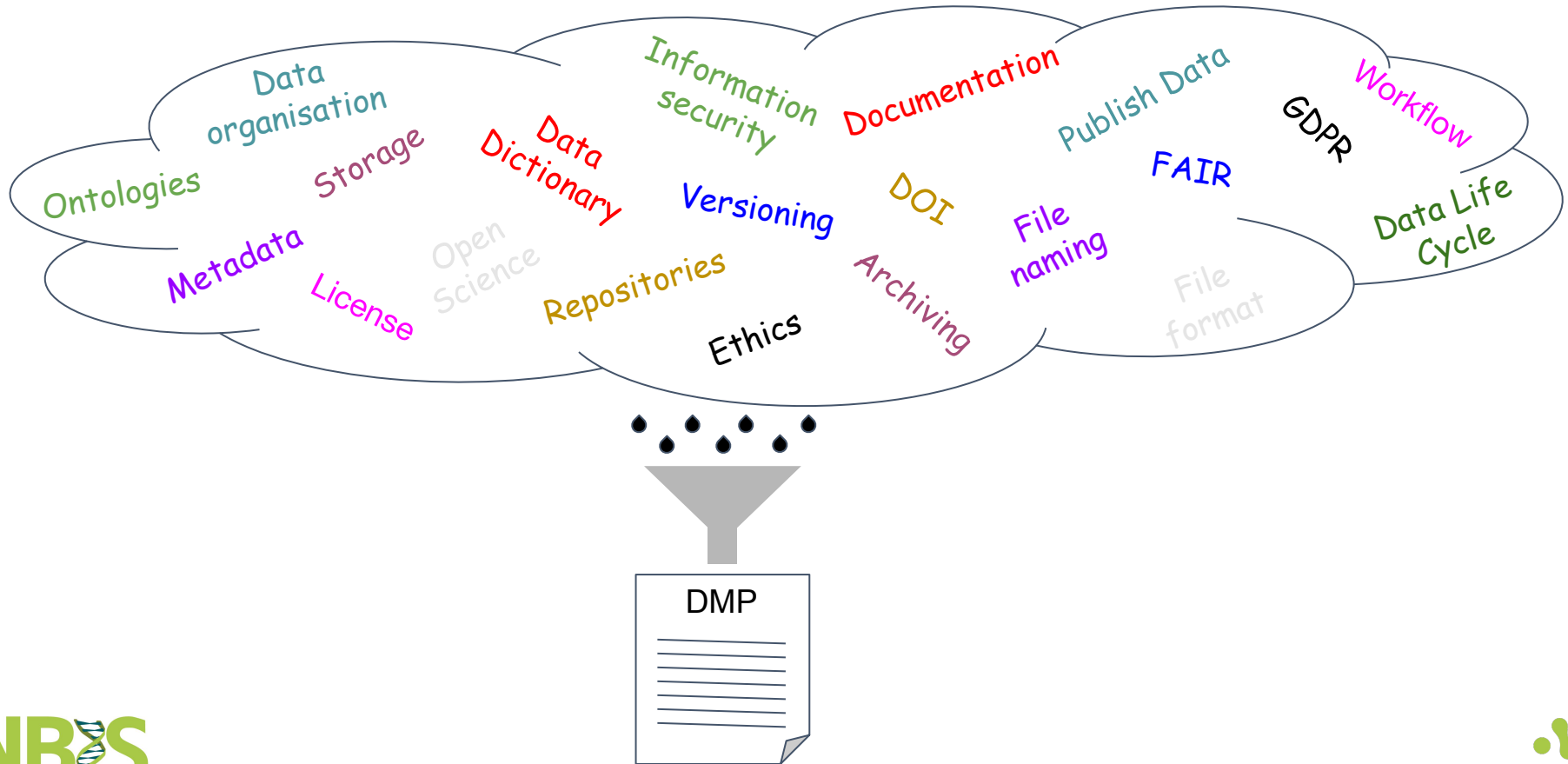
NBIS DM Team

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Learning objectives

1. Know the benefits of DMPs
2. Know the sections of a typical DMP
3. Know tools to assist in writing a DMP
4. Have experience in writing a DMP using an online tool called Data Stewardship Wizard
5. Know where to get help in future DMP writing

Data Management - Now what?



What is a DMP?

A document **addressing requirements and practices** for managing the project's data, code and documentation **throughout the data life cycle**.

Outlines the data management **strategies**, and **how** the data is:

- collected
- documented
- organised
- preserved



Data Life Cycle by [RDMkit](#) used under [CC-BY](#)



Why should you write a DMP?

- **Identify gaps** in data management
- **Define responsibilities**
- **Estimate costs**
- **Facilitate collaboration** by setting project wide standards (e.g. data organisation, naming conventions, metadata standards)



Data Life Cycle by [RDMkit](#) used under [CC-BY](#)

Why should you write a DMP?

- Obtain **well-managed** research data (which enables reproducibility and reusability)
- First step towards **FAIRness**
- **Requirement** from funders and other stakeholders for reasons of **transparency, openness and return on investment**



Data Life Cycle by [RDMkit](#) used under [CC-BY](#)

When do you write a DMP?

A DMP is a **living document** that will develop throughout the project

- **Project planning:** Outline the strategies to be able to estimate the resources needed, so this can be included in the proposal for funding.
- **Project start:** Complete with details e.g. about documentation, data quality measures, file and folder strategies, etc.
- **Project end:** Update with e.g. links to published data and details about archiving (what data and where).

The main parts of a DMP

1. Description of data

- What types of data will be created and/or collected?

Formats

Amount/volume of data

Instrument

Equipment

2. Documentation

- How will the material be documented and described?

ELN & LIMS

Collection method

Metadata standards

Versioning

The main parts of a DMP

3. Storage and backup

- How is data security, storage and backup handled?

Organization

Naming convention

Backup strategy

Security

4. Legal and ethical aspects

- How is data handled? Any legal requirements?

Sensitive data

GDPR

Intellectual property rights

Nagoya

The main parts of a DMP

5. Accessibility and long-term storage

- How, when, and where will research data and metadata be made accessible?

Repositories

Raw data

Code & Software

Type of storage

6. Responsibility and resources

- Who are the responsible persons for data management?

Resources

Labour estimate

The main parts of a DMP

1. Description of data

- What types of data will be created and/or collected, in terms of data format and amount/volume of data?

Reuse and
production of
research data

2. Documentation

- How will the material be documented and described, with associated metadata relating to structure, standards and format for descriptions of the content, collection method, etc.?

Documentation
procedures and
standards

3. Storage and backup

- How is data security, storage and backup of data and metadata safeguarded during the research process?

Storage,
security and
backup

cont. The main parts of a DMP

4. Legal and ethical aspects

- How is data handling according to legal requirements safeguarded, e.g. in terms of handling of personal data, confidentiality and intellectual property rights?

Legal & ethical
and assurance
procedures

5. Accessibility and long-term storage

- How, when and where will research data or information about data (i.e. metadata) be made accessible?
- In what way is long-term storage safeguarded, and by whom?

Future reuse
and long-term
access aspects

6. Responsibility and resources

- Who are the responsible persons for data management?
- What resources (costs, labour input or other) will be required for data management?

Responsibilities
and resources
required

How write a DMP?

DMP templates:

- Provided by funding agencies, e.g. [Swedish Research Council](#) and [Science Europe](#)
- High-level questions, with no guidance on how to answer
- Use MS Word?

DMP tools:

- [DMPOnline](#) - Common for universities; Good guidance but not Life Science specific; Free text answer type;
- [Data Stewardship Wizard](#) - Provided by [SciLifeLab](#); Life Science specific guidance; Many questions with answer options



Key Points

- A data management plan (DMP) is a document that describes the data produced in the course of a research project
- A DMP allows for well-managed data, and funding agencies often require a DMP for transparency and return on investment
- A DMP is a living document, the first version is written during project planning, and is then updated as the project proceeds
- There are standard templates available e.g. at funder agencies, and tools to assist when writing

Discuss your experience

1. Have you learned something today that you wish you knew earlier?
2. What is the biggest challenge you foresee in implementing a data management plan?
3. What is one question you still have about data management plans?